高硬钢用2刃微小径圆鼻铣刀
Hardened Steels 2-Flute Micro Diameter Corner Radius Endmills

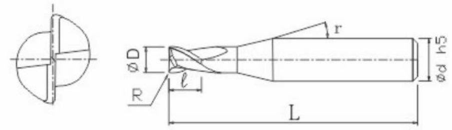
2刃 · 带R角 2-Flute · Corner Radius

平面 Planing · 侧面 Side Cutting · 槽 Slotting · 刻模 Die-sinking · 曲面 Profiling · R Radius · 螺旋 Helical · 粗加工 Roughing · 半精加工 Semi Finishing · 精加工 Finishing · 切削条件表 B016 Cutting Conditions

外径公差 Dia.tolerance: D≤0.1 : 0~- 0.005; 0.15≤D≤0.9 : 0~- 0.007 | r公差 r tolerance: ±0.005 | h5: 0~-0.005 (mm)

高硬度材料加工用。超群的切削排出性能，可实现高效率加工。
For machining of high-hardness materials. Excellent chip removal performance enables high efficiency.

TYPE A:



- 在高硬度领域(48-65HRC)的长时间加工中也可实现稳定的长寿命。
- 增大了螺旋角，使得刀刃变得更锋利，而且圆鼻形状可以抑制崩刃。
- Realized stably long tool life against high hardened steels (48~65HRC).
- Corner radius to prevent flute chipping, and high helix angle for cutting sharpness.

硬质合金 · X6m涂层 · 65HRC · R刃 · 螺旋角35° · 2

产品代码 Code No.	规格型号 Spec Typ.	外径(D) Dia.	角半径(R) Corner Radius	刃长(ℓ) Length of Cut	颈角(r) Neck Taper Angle	柄径(d) Shank Dia.	全长(L) Overall Length	形状 Type	定价 Retail Price
FX6MSR01001	0.1R0.01x0.2L	0.1	R0.01	0.2	15°	4	45	A	
FX6MSR01002	0.1R0.02x0.2L	0.1	R0.02	0.2	15°	4	45	A	
FX6MSR015002	0.15R0.02x0.3L	0.15	R0.02	0.3	15°	4	45	A	
FX6MSR015003	0.15R0.05x0.3L	0.15	R0.05	0.3	15°	4	45	A	
FX6MSR02002	0.2R0.02x0.4L	0.2	R0.02	0.4	15°	4	45	A	
FX6MSR02005	0.2R0.05x0.4L	0.2	R0.05	0.4	15°	4	45	A	
FX6MSR03005	0.3R0.05x0.6L	0.3	R0.05	0.6	15°	4	45	A	
FX6MSR0301	0.3R0.1x0.6L	0.3	R0.1	0.6	15°	4	45	A	
FX6MSR04005	0.4R0.05x0.8L	0.4	R0.05	0.8	15°	4	45	A	
FX6MSR0401	0.4R0.1x0.8L	0.4	R0.1	0.8	15°	4	45	A	
FX6MSR05005	0.5R0.05x1.0L	0.5	R0.05	1	15°	4	45	A	
FX6MSR0501	0.5R0.1x1.0L	0.5	R0.1	1	15°	4	45	A	
FX6MSR06005	0.6R0.05x1.2L	0.6	R0.05	1.2	15°	4	45	A	
FX6MSR0601	0.6R0.1x1.2L	0.6	R0.1	1.2	15°	4	45	A	
FX6MSR07005	0.7R0.05x1.4L	0.7	R0.05	1.4	15°	4	45	A	
FX6MSR0701	0.7R0.1x1.4L	0.7	R0.1	14	15°	4	45	A	
FX6MSR08005	0.8R0.05x1.6L	0.8	R0.05	1.6	15°	4	45	A	
FX6MSR0801	0.8R0.1x1.6L	0.8	R0.1	1.6	15°	4	45	A	
FX6MSR0802	0.8R0.2x1.6L	0.8	R0.2	1.6	15°	4	45	A	
FX6MSR9005	0.9R0.05x1.8L	0.9	R0.05	1.8	15°	4	45	A	
FX6MSR00901	0.9R0.1x1.8L	0.9	R0.1	1.8	15°	4	45	A	

切削参数参考表
Recommended Milling Conditions

加工材料 Work Material		淬火钢 Hardened Steels HPM-38 · STAVAX · SKD61 (~55HRC)				碳素钢 · 调质钢 Carbon Steels · Prehardened Steels S50C · NAK55 · NAK80 · HPM-1 (~43HRC)				铜 · 铝合金 Copper · Aluminum Alloy			
外径 Dia.	角半径(R) Corner Radius	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
0.1	0.01 0.02	30,000	50	0.001	0.008	30,000	100	0.005	0.008	30,000	150	0.008	0.008
0.15	0.02 0.05	30,000	100	0.002	0.02	30,000	150	0.008	0.02	30,000	200	0.01	0.03
0.2	0.02 0.05	30,000	150	0.003	0.04	30,000	200	0.01	0.05	30,000	250	0.012	0.05
0.3	0.05 0.1	30,000	180	0.003	0.08	30,000	200	0.02	0.1	30,000	300	0.024	0.1
0.4	0.05 0.1	30,000	300	0.005	0.1	30,000	350	0.025	0.12	30,000	450	0.03	0.12
0.5	0.05 0.1	25,000	400	0.01	0.12	30,000	500	0.03	0.14	30,000	650	0.036	0.14
0.6	0.05 0.1	25,000	400	0.02	0.13	30,000	600	0.035	0.16	30,000	800	0.04	0.16
0.7	0.05 0.1	30,000	500	0.03	0.16	30,000	700	0.04	0.2	30,000	1,000	0.05	0.2
0.8	0.1 0.2	25,000	10000	0.035	0.2	30,000	1,400	0.055	0.25	30,000	1,800	0.065	0.25
0.9	0.05 0.1	20,000	700	0.03	0.24	25,000	1,100	0.05	0.3	30,000	1,500	0.06	0.3

备注 Notes

※请根据实际的加工形状及使用机床等调整切削参。
※切深量的ap表示轴向切入量，ae表示步距量。
※加工淬火钢时，建议使用油雾冷却方式。
※轴向进刀建议采用螺旋进刀及倾斜进刀方式。
※沟槽切削时建议参考切削参数表，切深量ap及进给速度设定为50%以下，采用来回切削加工方式。
※发生振刀时，请以相同的比率降低主轴转速和进给速度。此外，主轴转速过低时，也以相同的比率调。
※Adjust milling conditions according to milling shape and machine type.
※ap : Axial Depth of Cut, ae : Radial Depth of Cut.
※Recommend to use oil mist coolant for machining hardened steels.
※Recommend to apply helical or ramping for approaching into axial direction.
※For slotting, recommend reciprocating milling by adjusting feed & ap in below 50% of recommended milling condition.
※Reduce both spindle speed and feed at same rate for chattering and also for insuicient spindle speed of a machine.